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Building the World's First Open-Source Multimodal SpeedRun

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Co-lead of the Kubernetes Policy WG, project lead of CNCF Security SIG, founder of the OpenStack Cyborg project, and co-founder of OpenStack Public Cloud WG.

Keynote Speech at various flagship open source summit such as OpenStack Summit, KubeCon and RISC-V Workshop

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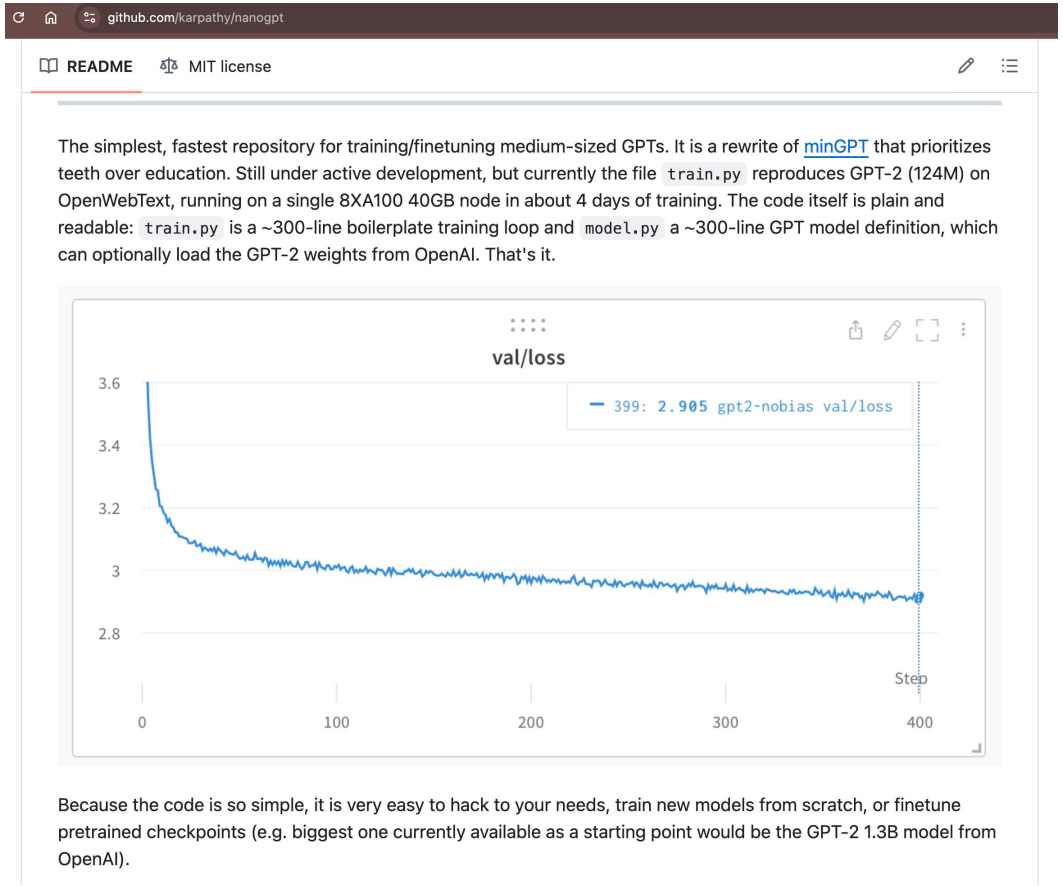
01 Background

- Metrics
  - Old era: best final score
  - New era: fastest path to target quality
  - Metrics: time-to-loss, samples-to-quality, cost-to-target

Benchmarks
- ssGPT-style small model
- Target val loss $\leqslant$ 3.28
- Under 90s on 8xH100 (repo claim)
- Under 400M tokens

Culture
- Public leaderboard
- Open PR optimization race
- Tiny gains celebrated



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Modded-NanoGPT

This repository hosts the *NanoGPT speedrun*, in which we (collaboratively|competitively) search for the fastest algorithm to use 8 NVIDIA H100 GPUs to train a language model that attains 3.28 cross-entropy loss on the FineWeb validation set.

The target (3.28 validation loss on FineWeb) follows Andrej Karpathy's GPT-2 replication in llm.c, which attains that loss after running for 45 minutes. The speedrun code also descends from llm.c's PyTorch trainer, which itself descends from NanoGPT, hence the name of the repo. Thanks to the efforts of many contributors, this repo now contains a training algorithm which attains the target performance in:

- Under 90 seconds on 8xH100 (the llm.c GPT-2 replication needed 45 minutes)
- under 400M tokens (the llm.c GPT-2 replication needed 10B)

This improvement in training speed has been brought about by the following techniques:

- Modernized architecture: Rotary embeddings, QK-Norm, and ReLU²
- The Muon optimizer [writeup] [repo]
- Use FP8 matmul for head, and asymmetric rescale and softcap logits
- Initialization of projections to zero (muP-like)
- Skip connections from embedding to every block as well as from block 3 to 6
- Extra embeddings which are mixed into the values in attention layers (inspired by Zhou et al. 2024)
- Flash Attention 3 with long-short sliding window attention pattern (inspired by Gemma 2) and window size warmup with YaRN
- Align training batch starts with EoS and set a max document length
- Accumulate gradients for 2 steps for embedding and lm_head before updating parameters
- Backout, with single activation input for last 3 attention layers
- Polar Express implementation in Muon
- Smear module to enable 1 token look back
- Sparse attention gate
- NorMuon
- Cautious Weight Decay w/ schedule tied to LR
- Exponential decay of residual stream
- Batch size schedule
- Max seq length schedule
- Partial Key Offset

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	minutes	fuse fp8 quantization in LM head			
68	1.535 minutes	Move bigram hash to GPU	01/31/26	log,PR	@dhrvji
69	1.528 minutes	Kernel Optimizations	02/02/26	log,PR	@EmmettBicker & AI System Aster
70	1.521 minutes	Tune value embed layout and ve_gates	02/03/26	log,PR	@photon_mz
71	1.516 minutes	Sparse bigram gradient comms and optimized loading on CPU	02/06/26	log,PR	@roeeshenberg
72	1.496 minutes	Increase minimum lr and add max_seq_len schedule	02/10/26	log,PR	@dualverse-ai & AI System Station
73	1.485 minutes	Partitioned Hyperconnections	02/12/26	log,PR	@sisovicm
74	1.468 minutes	Flattened GPT forward, removed post attention lambdas, added transpose kernels	02/16/26	log,PR	@ChrisJMcCormick
75	1.453 minutes	Cross Entropy Kernel Optimizations	02/23/26	log,PR	@moof2x
76	1.446 minutes	Reuse and tune backward transpose kernel	02/28/26	log,PR	@samacqua
77	1.435 minutes	Replace partitioned hyperconnections with single saved activation	03/06/26	log,PR	@classiclarryd
78	1.426 minutes	Tighten bounds on fa3 max_num_docs to match fineweb distribution	03/22/26	log,PR	@ChrisJMcCormick
79	1.411 minutes	Fuse Cross Entropy Fwd/Bwk Kernel, to avoid recalc on softcap sigmoid	04/04/26	log,PR	@andrewbriand8
80	1.406 minutes	In Muon orthogonize Q and K matrices in pairs of heads, instead of across the full 6 head matrix	04/08/26	log,PR	samacqua

Speedrun track 2: GPT-2 Medium

The target loss for this track is lowered from 3.28 to 2.92, as per Andrej Karpathy's 350M-parameter llm.c baseline. This baseline generates a model with performance similar to the original GPT-2 Medium, whereas the first track's baseline generates a model on par with GPT-2 Small. All other rules remain the same.

> Note: `torch._inductor.config.coordinate_descent_tuning` is turned on after the record 6 (*).

#	Record time	Description	Date	Log	Contributors
1	5.8 hours	llm.c baseline (350M parameters)	05/28/24	log	@karpathy, llm.c contributors
2	29.3 minutes	Initial record based on scaling up the GPT-2 small track speedrun	01/18/25	log	@kellerjordan0
3	28.1 minutes	Added standard weight decay	02/08/25	log	@kellerjordan0
4	27.7 minutes	Tuned Muon Newton-Schulz coefficients	02/14/25	log	@leloykun
5	27.2 minutes	Increased learning rate cooldown phase duration	03/06/25	log	@YouJiacheng
6	25.95 minutes*	2x MLP wd, qkv norm, all_reduce/opt.step() overlap, optimized skip pattern	03/25/25	log	@YouJiacheng
7	25.29 minutes	Remove FP8 head; ISRU logits softcap; New sharded mixed precision Muon; merge weights	04/16/25	log	@YouJiacheng
8	24.50 minutes	Cubic sliding window size schedule, 2× max window size (24.84 minutes) 24.5min repro	04/22/25	log	@jadenj3o
9	24.12 minutes	Add two value embeddings	08/28/25	log, PR	@snimu
10	24.07 minutes	Second input embedding	09/11/25	log, PR	@snimu
11	23.45 minutes	Upgrade from torch 2.7 to torch==2.10.0.dev20251210+cu126	-	-	-

Modded-NanoGPT Optimization Benchmark

The goal of this benchmark is to collaboratively|competitively find efficient neural network optimizers. Unlike the main NanoGPT speedrun which seeks to minimize *wallclock time* by any means, here we aim to minimize *step count* by improving the optimization algorithm (⇒ methods that are slow in terms of wallclock are perfectly OK).

Quickstart

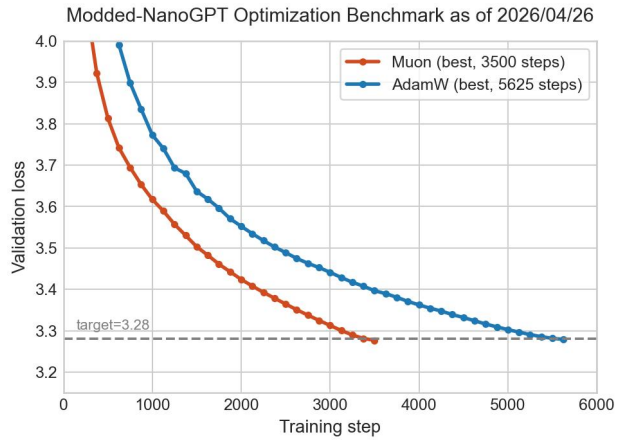
The baseline can be run using the following command on any {1,2,4,8}x-{A,H}100 machine:

```
git clone https://github.com/KellerJordan/modded-nanogpt.git && cd modded-nanogpt
pip install torch==2.10 huggingface_hub
python data/cached_fineweb10B.py 40  # downloads 4B training tokens
torchrun --standalone --nproc_per_node=$(nvidia-smi -L | wc -l) records/track_3_optimization/train_gpt_simple.py
```

Notable results history

The following results each improved the best known hyperparameters for an optimizer on this benchmark. Many more non-SOTA results (e.g., from hyperparameter sweeps) can be found in `results/`.

#	Steps to 3.28	Description	Date	Log	Contributors
1	3600	Muon lr=.02 wd=.01	2026/04/26	log	@kellerjordan0
2	5625	AdamW lr=0.0015 wd=0.1 betas=(0.9, 0.95) warmup_steps=250	2026/04/26	log	@kellerjordan0
3	3500	Muon lr=.025 wd=.0125	2026/04/26	log	@kellerjordan0



- What is Marin
  - Open training research platform
  - Reproducibility-first
  - Shared infrastructure
  - Community experiments

Difference
  - Not a single WR race
  - Supports many future speedruns on systemic level
  - Vairous size e2e training runs

Marin

Developing models together, openly

GitHub
Discord
Mailing list
Documentation
DeepWiki
HuggingFace
WandB
Speedrun Leaderboard

We also publish reusable analysis views, including the Perplexity Gap Dashboard.

Some examples:

1. Experiment 935: How does z-loss impact loss? [issue, PR, code, execution, WandB]
2. Experiment 950: How does pretraining learning rate impact SFT? [issue, PR, code, execution, WandB]
3. Experiment 163: Is BERT a better quality filter than fastText? [issue, PR, code, execution, WandB]
4. Experiment 1290: Which optimizers actually outperform AdamW? [issue, PR, code, execution, WandB]
5. Experiment 1183: Are MoEs really better than dense models? [issue, PR, code, execution, WandB]
6. Experiment 702: How should you train on rare task-relevant data? [issue, PR, code, execution, WandB]

Models 🌐

We trained some models in Marin:

1. Marin-8B-Base (deeper-starling) [issue, code, execution, WandB]: Beats Llama 3.1 8B base on 14/19 standard benchmarks! Read more in our retrospective.
2. Marin-8B-Instruct (deeper-starling-05-15) [execution]: Try it out on Together AI!
3. Marin-32B-Base [issue, code, execution]: Beats OLMo 2 32B Base on 14/19 standard benchmarks (making it the best open-source model as of Oct 29, 2025), and is close to Gemma 3 27B PT and Qwen 2.5 32B Base (the best comparably-sized open-weight models). Read more in our retrospective.

Speedrun 🏃

Have a new architecture or training procedure that you think is more efficient? Participate in the Marin speedrun competition (inspired by the nanogpt speedrun), pick your compute budget, and create the fastest method to train a model to a certain quality! Here's a tutorial for running your first speedrun on GPUs. We will offer free compute to scale up top performers. Get started here.

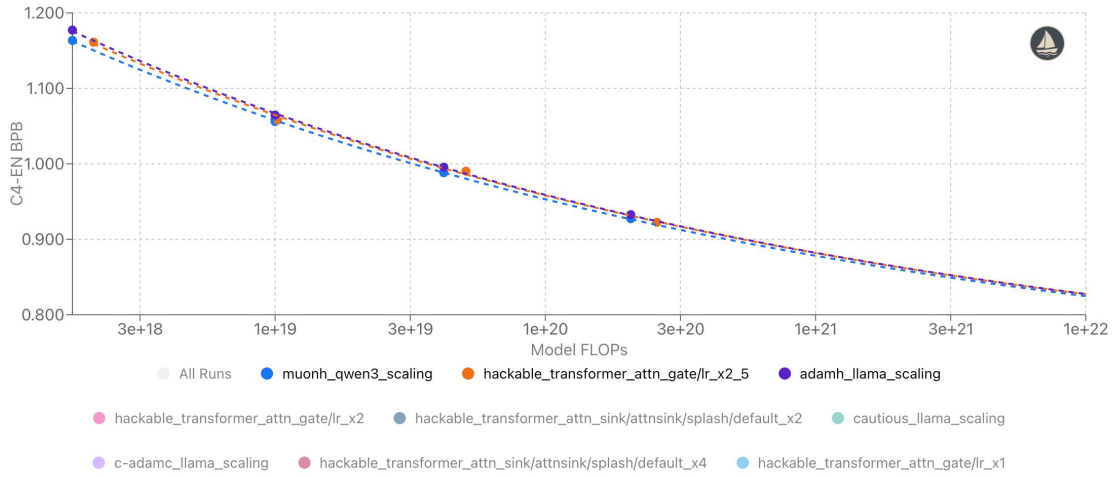
Total Runs in Track	Best Compute Scaling Term	Best Projected BPB @ 1e+22 FLOPs
23	-0.1657	0.8247

Performance Chart

Download Chart

X-axis metric: ○ Training Hardware FLOPs ◉ Model FLOPs

Y-axis metric: ◉ Absolute BPB ○ Relative to Baseline



Scaling Leaderboard

Fit on `exp(intercept) × compute^slope + asymptote`, ranked by projected BPB @ 1e22 FLOPs

RUN NAME	AUTHOR	DATE ADDED	INTERCEPT	SLOPE	ASYMPTOTE	R^2	PROJECTED BPB	W&B RUN
muonh_qwen3_scaling	Kaiyue Wen Stanford University	2025-10-11	5.364	-0.146	0.690	1.000	0.825	View Rur
hackable_transformer_attn_gate/lr_x2_5	Calvin Xu Stanford University	2026-02-02	5.359	-0.145	0.689	0.998	0.827	View Rur
adamh_llama_scaling	Kaiyue Wen Stanford University	2025-10-10	5.496	-0.148	0.693	1.000	0.828	View Rur
hackable_transformer_attn_gate/lr_x2	Calvin Xu Stanford University	2026-02-02	5.551	-0.150	0.702	0.998	0.829	View Rur
hackable_transformer_attn_sink/attnsink/splash/default_x2	Calvin Xu Stanford University	2025-11-13	5.586	-0.151	0.702	0.999	0.830	View Rur
cautious_llama_scaling	William Held Georgia Tech	2025-07-03	5.732	-0.154	0.702	0.999	0.830	View Rur
c-adamc_llama_scaling	William Held Georgia Tech	2025-07-03	5.772	-0.155	0.705	0.999	0.831	View Rur
hackable_transformer_attn_sink/attnsink/splash/default_x4	Calvin Xu Stanford University	2025-11-15	5.229	-0.141	0.682	0.996	0.833	View Rur
hackable_transformer_attn_gate/lr_x1	Calvin Xu Stanford University	2025-12-06	5.793	-0.157	0.718	0.998	0.834	View Rur

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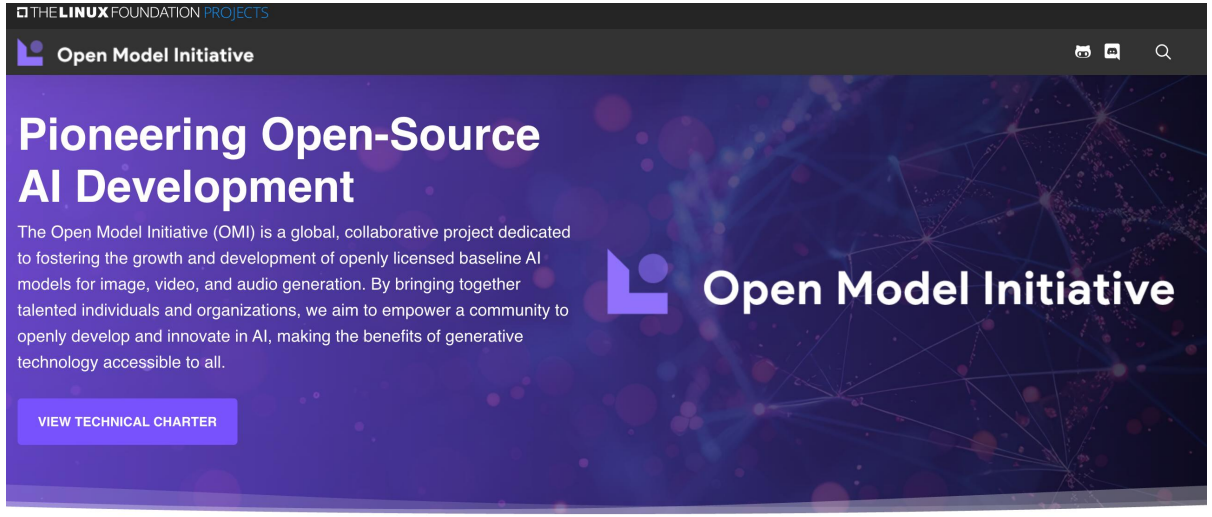
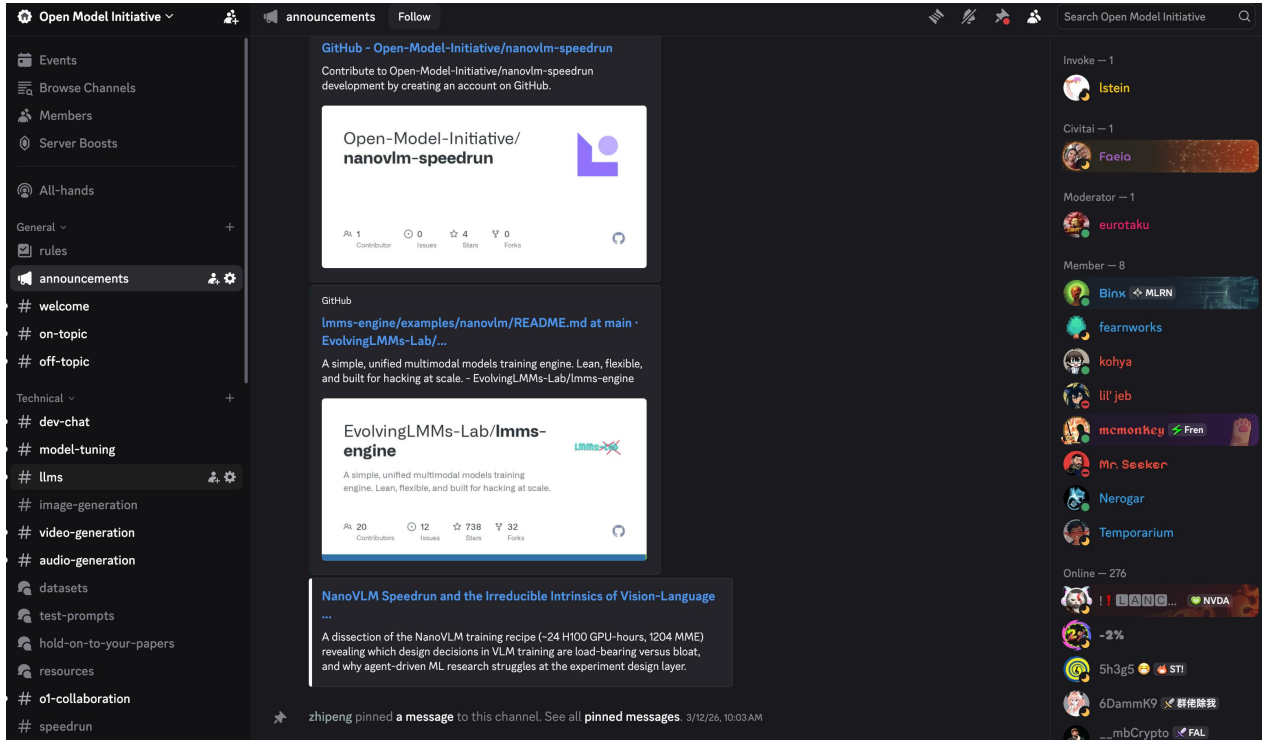
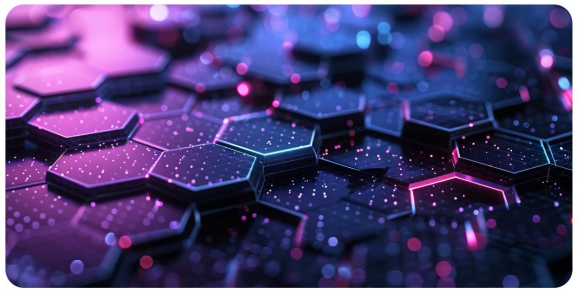




Our Vision and Mission

The Open Model Initiative is driven by a commitment to openness and collaboration. Our mission is to:

- Create and Maintain Openly Licensed AI Models
- Promote Responsible AI Development and Transparency
- Support Cross-Community and Cross-Project Collaboration



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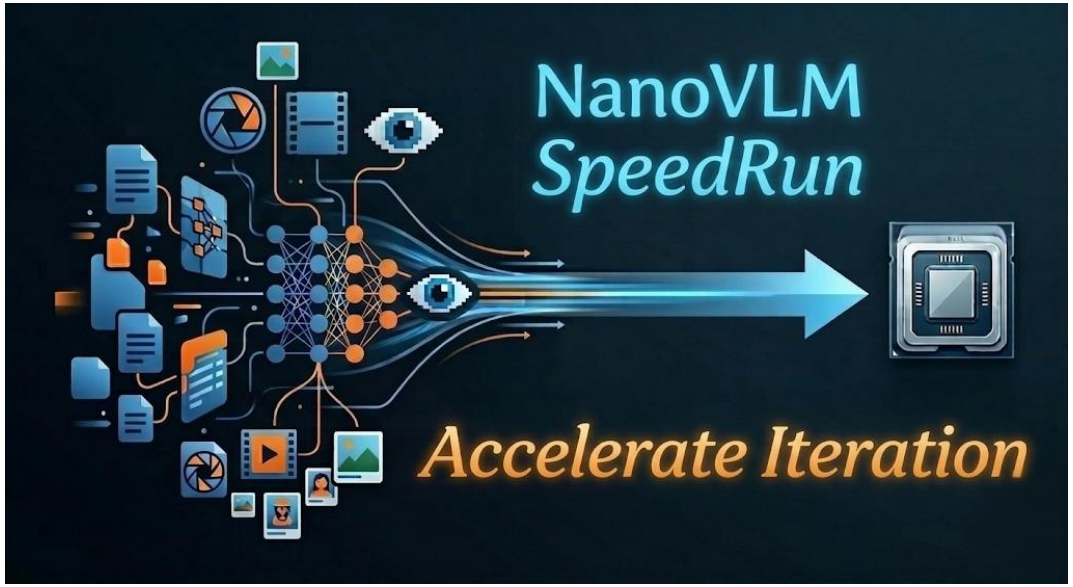
03

Multi-Modal SpeedRun

Define Problems, Set Rules, Build Baselines, and Collaborate on Benchmarking

https://github.com/Open-Model-Initiative/imagegen-speedrun/



https://github.com/Open-Model-Initiative/nanovlm-speedrun

imagegen-speedrun / speedrun-basic-rule-set.md

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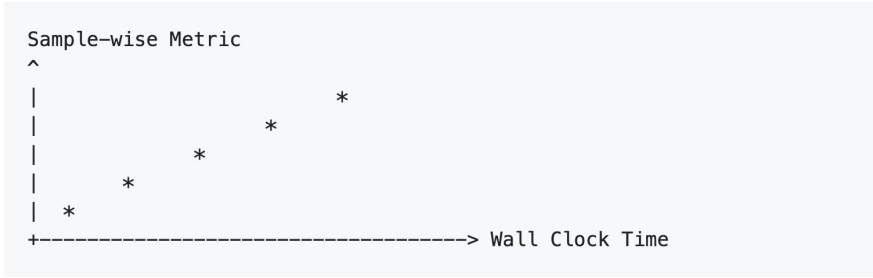
Primary Evaluation Curve

Each method must report:



- **x-axis**: Wall Clock Time
- **y-axis**: Sample-wise t2i metrics (GenEval1/2, HPSv3)

Note:

- The metric is open to discussion.

3. Dataset and Task

Item	Specification
Dataset	ImageNet (train split)
Resolution	256*256 resolution
Sampling steps (NFE)	100 (fixed across runs)

2821c4b SpeedRun-DiT-202601 / results.txt

SwayStar123 add xl results

Code Blame 460 lines (374 loc) · 15.1 KB

```
XL/1 @ 100k 2.99 it/s
Inception Score: 122.14485168457031
FID: 7.156954631595056
sFID: 4.976955193591721
Precision: 0.75316
Recall: 0.5972

XL/1 @ 400k 2.99 it/s
Inception Score: 178.141357421875
FID: 2.996033075112507
sFID: 4.411010295598999
Precision: 0.76946
Recall: 0.6317

    with SPRINT + RMS + ROPE + VALRES:
        Inception Score: 279.7873229980469
        FID: 3.101697263426729
        sFID: 4.9480099019050385
        Precision: 0.83302
        Recall: 0.5651

        kd_value: 0.24849
        kd_variance: 0.00434

        with RELU2:
            Inception Score: 265.0180969238281
            FID: 2.61406202232871
            sFID: 4.812557585819945
            Precision: 0.81848
            Recall: 0.5853
```

Primary Leaderboard

Each method must report a single leaderboard entry that includes both benchmark scores and training compute.

For the initial NanoVLM release, the leaderboard should contain:

Method	Train Time	Total FLOPs	MME	MMMU_Val	MMStar	GQA	ChartQA	DocVQ
NanoVLM Baseline	23.75 GPU hours (8 x H100)	335.02 PFLOPS	1204.46 (948.75/255.71)	0.3022	0.3273	0.4184	0.1084	0.1018

Reported benchmark metrics include:

- `MME` (Perception/Cognition)
- `MMBench` (EN Dev)
- `MMMU_Val`
- `MMStar` (Average)
- `GQA` (Exact Match)
- `ChartQA` (Relaxed Overall)
- `DocVQA` (ANLS)
- `OCRBench`
- `POPE` (Accuracy)

Item	Specification
Stage 1 data	`LMMs-Lab-Speedrun/Data_NanoVLM/Stage1-LLaVA-Pretrain`
Stage 2 data	`LMMs-Lab-Speedrun/Data_NanoVLM/Stage2-LLaVA-NeXT-Data`

The initial task is to train a compact VLM with the released two-stage recipe:

- **Stage 1**: projector-only alignment between vision and language
- **Stage 2**: instruction tuning on the released multimodal data recipe

Note:

- Local YAML path configuration is allowed, but the underlying dataset content and split policy should remain unchanged.
- Any major dataset change should be proposed as a separate track rather than silently changing the baseline.

Baseline Recipe

The initial NanoVLM SpeedRun baseline uses:

- **LLM**: `Qwen/Qwen3-0.6B`
- **Vision Encoder**: `google/siglip2-so400m-patch16-naflex`
- **Projector**: LLaVA-style 2-layer MLP
- **Training Paradigm**: 2-stage training

Model Changes

- The baseline recipe is the default comparison point
- Participants may propose architecture or training changes
- Any deviation from the baseline recipe must be clearly reported
- Total trainable parameter count must be reported

Note:

- Future versions may split submissions into a strict baseline track and a more open innovation track.

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Stage	Frozen	LR	Batch	Steps	PFLOPS	H100-hrs
1 (Alignment)	vision + LLM	10^{-3}	32×8	2180	236.79	19.32
2 (Instruction)	vision only	2×10^{-5}	8×8	11540	98.23	4.43

Total: ~335 PFLOPS, ~24 H100-hours. MME: 1204.46. Data: 558K caption pairs (Stage 1), 738K filtered instruction samples (Stage 2).

Training Intrinsics

Intrinsic 1: The Vision Encoder Stays Frozen

SigLIP2-so400m stays frozen throughout both stages. This follows the standard LLaVA two-stage recipe: freeze the vision encoder, train only the projector (Stage 1), then unfreeze the LLM while keeping the vision encoder frozen (Stage 2). The practical motivation is straightforward - the vision encoder is already a strong feature extractor out of the box, and fine-tuning it on 558K captioning pairs risks degrading features learned from much larger pretraining data, while also increasing GPU memory requirements significantly.

Whether this is strictly necessary is an open question. Some VLM recipes (e.g., PaLI, InternVL) do fine-tune the vision encoder at larger data scales and report gains. At the NanoVLM compute budget, freezing is the conservative and validated choice.

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The Agent Research Gap

Karpathy recently observed that AI agents running nanochat optimization experiments produce poor experiment designs. They vary parameters without controlling for compute, generate spurious results (e.g., "discovering" that larger hidden dimensions lower validation loss - trivially true in the infinite-data regime), and fail to create strong baselines before ablating.

NanoVLM illustrates exactly what makes this hard. The recipe above contains dozens of coupled decisions - which components to freeze, learning rate ratios, batch sizes per stage, data filtering thresholds. Each decision interacts with every other. An agent that independently varies hidden dimension will produce results confounded by uncontrolled variables.

The deeper problem: training intrinsics are not in the loss landscape. You cannot gradient-descend your way to "freeze the vision encoder." You need meta-knowledge that vision encoders are already well-trained and that 558K samples cannot improve them. This is the kind of tacit knowledge that Karpathy's nanochat miniseries makes explicit through iso-FLOP methodology - always compare at fixed compute, use task-level metrics (CORE) not raw validation loss.

Formalizing the Design Space

Can we make training intrinsics navigable? Consider a decision DAG:

```
vision_encoder -> freeze_strategy -> projector_design -> stage1_lr
              \-> data_budget -> stage1_data (alignment)
                             \-> stage2_data (instruction, filtered)
              \-> compute_budget -> batch_per_stage -> memory_constraint
```

Each node has a small set of valid configurations. Edges encode constraints (if vision encoder is frozen, projector must be trainable). An agent navigating this DAG with iso-FLOP baselines at each node could systematically discover NanoVLM-quality recipes.

The question is whether this DAG can be learned from the literature, or whether it requires the kind of judgment that accumulates from years of failed training runs.

04

Future

LFAI&Data Foundation/Open Model Initiative
https://github.com/Open-Model-Initiative/

Nanovlm SpeedRun
https://github.com/Open-Model-Initiative/nanovlm-speedrun

SpeedRun AI-CoScientist
https://coscientist.lmms-lab.com/p/cs/qLTGKSaKjCcnoB7zI_bcU

ImageGen SpeedRun
https://github.com/Open-Model-Initiative/imagegen-speedrun

Omni-Claw
https://github.com/Open-Model-Initiative/omni-claw

Omni-RLM
https://github.com/Open-Model-Initiative/Omni-RLM

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Thanks!